

CONTEXT-CONDITIONED SEQUENTIAL KEYPOINT DENSITY ESTIMATION FOR TRAFFIC HAZARD DETECTION

PROGRESS REPORT

Doruk Topcu

Department of Computer Engineering
Hacettepe University
doruktopcu@hacettepe.edu.tr
N25142281

ABSTRACT

The Sequential Keypoint Density Estimator (SeeKer / SKDE) is a recent ICCV 2025 baseline for skeleton-based video anomaly detection that models the joint density of a pedestrian’s 2D keypoint trajectory through an autoregressive factorization. Despite its simplicity, SeeKer matches or surpasses much heavier graph-convolutional and diffusion-based competitors. However, it is *environmentally blind*: each skeleton is treated in complete isolation from the surrounding scene. In real-world surveillance — and especially in traffic-hazard scenarios — a pedestrian’s motion is rarely abnormal in absolute terms; it is the *interaction* with vehicles, bicycles, or restricted zones that defines the hazard. We propose **CCSKDE** (Context-Conditioned SKDE), an extension that injects a lightweight environmental context vector C_t — derived from off-the-shelf YOLOv11 detections of non-pedestrian scene elements — directly into the autoregressive factorization. We evaluate on the ShanghaiTech Campus dataset, which natively contains pedestrian–vehicle interactions. This document reports our problem framing, related work, the proposed method, the experimental protocol, and preliminary results from reproducing the SeeKer baseline.

1 PROBLEM DEFINITION

Video anomaly detection (VAD) is the task of identifying frames or events in a video stream whose appearance, motion, or interaction patterns deviate significantly from those observed in a training set of *normal* videos. Modern VAD pipelines increasingly operate not on raw RGB pixels but on intermediate, privacy-preserving representations such as 2D human skeletons extracted by an off-the-shelf pose estimator. Skeleton-based VAD has three practical advantages: (i) the input is much lower-dimensional than RGB, (ii) models become invariant to appearance and lighting biases that often plague RGB-based detectors, and (iii) no facial or identifying pixels are retained, which is desirable for surveillance applications.

Within this family, the *Sequential Keypoint Density Estimator* (SKDE / SeeKer) of Delić et al. (2025) is a particularly elegant baseline. It treats a pedestrian’s pose trajectory as a high-dimensional time series and estimates its joint density via an autoregressive factorization at the keypoint level: each individual 2D keypoint is predicted as a Gaussian conditioned on all previously generated keypoints in the same and earlier frames. Anomalies are detected by ranking observed test sequences by their log-likelihood under the model. Despite using only a small masked fully-connected (MADE) network (Germain et al., 2015), SeeKer matches or exceeds heavier flow-based and graph-based competitors on standard benchmarks (Hirschorn & Avidan, 2023; Liu et al., 2018).

Limitation we address. SeeKer’s autoregressive distribution is conditioned only on the pedestrian’s own past keypoints; it never observes *what else is in the scene*. This is a fundamental in-

formation bottleneck: a pedestrian walking at a constant velocity in a calm campus square and the same pedestrian walking at the same velocity directly into the path of a moving vehicle yield identical SeeKer likelihoods. In surveillance, however, the latter is precisely the event we need to flag. The aim of this project is to extend SeeKer with a minimal, mathematically faithful conditioning mechanism on a per-frame *environmental context vector* C_t , so that the predicted keypoint distribution can sharpen (smaller covariance) when the pedestrian is in a high-risk spatial configuration.

Challenges. The problem is non-trivial for four reasons. (i) **Unsupervised conditioning:** VAD is one-class, so we must learn what contextual configurations are normal without any hazard labels. (ii) **Preserving the autoregressive factorization:** naively concatenating C_t to MADE’s input breaks the carefully designed keypoint mask, so the conditioning must be injected as an unconditional shortcut while the keypoint-to-keypoint causality is left intact. (iii) **Generalization across scenes:** ShanghaiTech contains 13 different camera viewpoints, so C_t must be a scene-scale-invariant function of pedestrian–object geometry rather than absolute image coordinates. (iv) **Compute envelope:** the project must run in a single-semester timeline on commodity hardware (Apple Silicon MPS in our case), ruling out heavy dynamic-graph or transformer backbones. We note that ShanghaiTech is the only widely-used VAD benchmark with mixed pedestrian/vehicle interactions; UBnormal (Ac-sintoae et al., 2022) is fully synthetic, so our context contribution is evaluated primarily on ShanghaiTech.

2 RELATED WORK

Skeleton-based video anomaly detection. Pose-based approaches have become the dominant paradigm for VAD on person-centric datasets, since they sidestep RGB-level confounders. Morais et al. (2019) introduced one of the first skeleton-only VAD pipelines using a recurrent message-passing network. Markovitz et al. (2020) cast normal motion modeling as soft-clustering of spatio-temporal graph embeddings. STG-NF (Hirschorn & Avidan, 2023) fits a normalizing flow over spatio-temporal graphs of joints and was the state of the art on ShanghaiTech-HR until SeeKer. SeeKer (Delić et al., 2025) departs from these heavier architectures and demonstrates that an autoregressive masked fully-connected network is in fact a strong — and previously overlooked — baseline. Our work builds directly on SeeKer.

Density estimation backbones. SeeKer’s core is MADE (Germain et al., 2015), a masked fully-connected autoencoder that performs autoregressive density estimation in a single forward pass. Related neural density estimators (RealNVP (Dinh et al., 2017), Glow (Kingma & Dhariwal, 2018)) require multiple coupling layers but produce exact log-likelihoods; STG-NF exploits this for VAD scoring. SeeKer shows the simpler masked-FC formulation is sufficient and substantially cheaper.

Context-aware anomaly detection. A separate strand of VAD literature explicitly injects scene/object information into the anomaly scorer. Ionescu et al. (2019) score per-object appearance and motion features using one-class SVMs, and Georgescu et al. (2021) use background-aware self-supervised pretexts. The HSC (Sun & Gong, 2023) and SSMTL++ (Georgescu et al., 2022) frameworks combine multiple self-supervised signals including object-level cues. To our knowledge, no prior work has injected an explicit, per-frame proximity-to-non-pedestrian context vector into an autoregressive keypoint density estimator; this is the gap our work addresses.

Multi-agent and interaction anomalies. Closest in spirit are methods that model *interactions* among actors. Wiederer et al. (2022) learn a density over multiple road-agent trajectories and flag low-density configurations, and ComplexVAD (Doshi & Yilmaz, 2025) explicitly targets “complex” anomalies arising from the interaction of two or more objects (e.g. a cyclist colliding with a car). These operate on trajectory- or graph-level features. Our contribution is orthogonal in mechanism: we retain SeeKer’s autoregressive *keypoint-level* density but widen its support from a single human skeleton to a heterogeneous set of road agents, so that a pedestrian’s per-keypoint likelihood is conditioned on the surrounding vehicles within *one* factorised density. To our knowledge this is the first multi-agent extension of an autoregressive keypoint density estimator. We emphasise that the vehicle representation is an oriented bounding box rendered as keypoints, not an articulated skeleton (Section 3.2).

Object detection, datasets and pose extraction. We use YOLOv11 (Ultralytics, 2024) for one-time offline detection of non-pedestrian scene elements; its bounding boxes are aggregated into C_t as described in Section 3.2. ShanghaiTech Campus (Liu et al., 2018) comprises 13 outdoor scenes, 437 videos, and a mixture of pedestrian, bicycle, and vehicle activity; anomalies (bicycle riding in pedestrian zones, unauthorized vehicles, etc.) are typically defined by a pedestrian–object interaction, making it the natural benchmark for our hypothesis. All skeletons are AlphaPose (Fang et al., 2017) COCO-17 keypoints converted to COCO-18 (synthetic neck) following SeeKer’s preprocessing; we use the publicly released pose JSONs from STG-NF for full reproducibility.

3 METHOD

3.1 BASELINE: SEEKER IN DETAIL

Let a pedestrian track be a sequence of T frames, each containing $V = 18$ COCO-18 2D keypoints. We denote an individual coordinate by $X_{t,n} \in \mathbb{R}^2$, the n -th keypoint of frame t , with $n \in \{1, \dots, V\}$. Let $X_{<t}$ denote all keypoints from frames before t , and $X_{t,<n}$ all keypoints of frame t with index strictly less than n . SeeKer factorizes the joint density of the entire track as

$$p_\theta(X_{1:T}) = \prod_{t=1}^T \prod_{n=1}^V p_\theta(X_{t,n} \mid X_{t,<n}, X_{<t}). \quad (1)$$

Each conditional is parameterized as a 2D Gaussian $\mathcal{N}(\mu_\theta, \Sigma_\theta)$ whose mean and covariance are produced by a single masked fully-connected network (*MADEPartial*). Following SeeKer’s headline configuration — and the publicly released implementation we build on — the covariance is *diagonal*, $\Sigma_\theta = \text{diag}(\sigma_{\theta,x}^2, \sigma_{\theta,y}^2)$: the network emits, per keypoint, a mean and a log-variance for each coordinate, so the negative log-likelihood reduces to a per-coordinate Mahalanobis term plus a log-variance penalty. (SeeKer additionally *ablates* a full 2×2 covariance through a Cholesky factor $\Sigma_\theta^{-1} = L_\theta L_\theta^\top$, but reports it to perform on par with the diagonal form (77.8 vs. 77.9 on UBnormal); we therefore retain the diagonal parameterisation and instead make the variance *context-dependent*, Section 3.2.) The keypoint-level autoregressive constraint is enforced by binary masks on the weight matrices, exactly as in MADE (Germain et al., 2015); SeeKer adapts the masking so that conditioning is causal both *across* time and *within* the keypoint ordering of the current frame.

Training. The model is trained on *normal* pedestrian sequences only, by minimizing the negative log-likelihood of Equation 1.

Anomaly scoring. At inference, each test frame receives a per-keypoint negative log-likelihood. SeeKer aggregates these into a frame-level anomaly score (optionally weighted by AlphaPose detection confidence), then applies temporal Gaussian smoothing. The resulting score is the Mahalanobis distance under the predicted Σ_θ : when the network is confident (small Σ_θ) and the observation deviates from μ_θ , the score is large.

3.2 PROPOSED EXTENSION: CCSKDE

Our contribution is to make the predicted distribution *context-aware*, by introducing a per-frame context vector C_t :

$$p_\theta(X_{t,n} \mid X_{t,<n}, X_{<t}, C_t). \quad (2)$$

Constructing C_t . For each frame t in the training and test videos, we run YOLOv11 on the raw RGB frame and retain only detections of non-pedestrian classes that are plausible hazards on a campus: `car`, `truck`, `bus`, `motorcycle`, `bicycle`. For each such detection b , with bounding-box centre (x_b, y_b) , we compute its image-space distance to the target pedestrian’s skeleton centroid $\bar{x}_t$: $d_{t,b} = \|(x_b, y_b) - \bar{x}_t\|_2 / \sqrt{H^2 + W^2}$, normalized by the image diagonal so the quantity is scene-scale-independent. We aggregate to a fixed-dimension vector by taking, per class, the inverse minimum distance and the count of objects within a fixed radius:

$$C_t = [1/d_{c_1}^{\min}, n_{c_1}^{\leq r}, 1/d_{c_2}^{\min}, n_{c_2}^{\leq r}, \dots] \in \mathbb{R}^{2K}, \quad (3)$$

where K is the number of hazard classes. Frames with no relevant detection receive an all-zero vector. This design has three properties we care about: (i) it is fixed-dimension regardless of how many objects are in the scene, (ii) its magnitude grows monotonically with proximity to a hazard, and (iii) it is computed once offline, so it adds no train-time inference cost on the YOLO side.

Coordinate-frame consistency (a subtle but decisive detail). The proximity in Eq. 3 is only meaningful if the pedestrian centroid $\bar{x}_t$ and the object centres (x_b, y_b) live in the *same* coordinate frame. SeeKer normalises each pose segment to zero-mean / unit-variance before the network sees it, which destroys absolute image position; computing $\bar{x}_t$ from that normalised pose (as a naïve implementation does) makes C_t collapse to a pedestrian-*independent* function of where vehicles sit in the image. We instead compute $\bar{x}_t$ from the *raw* pixel keypoints divided by the frame size (W, H) , placing it in the same $[0, 1]$ image frame as the YOLO boxes. This single correction is what makes the conditioning carry genuine pedestrian–object geometry.

Oriented vehicle-keypoint context (the “car matrix”). The scalar proximity above discards the *geometry* of the hazard: its orientation, extent, and which side of the pedestrian it occupies. We therefore introduce a richer alternative in which each of the M nearest hazards is described by a compact *oriented keypoint set* — “like a bounding box, but better”. From a YOLOv1 l-segmentation mask we fit an oriented rectangle (`minAreaRect`) and read off $K_v = 6$ keypoints: the four oriented corners, the centre, and a heading point along the major axis. Each keypoint is expressed *relative* to the pedestrian centroid and divided by the pedestrian’s skeleton scale, so the representation is translation- and scale-invariant (directly addressing challenge iii). The resulting fixed-dimension “car matrix” $C_t \in \mathbb{R}^{M(2K_v+2)}$ (corners, centre, heading per vehicle, plus an inverse-distance and a class index) replaces the proximity vector while using the identical injection machinery below. We stress that this is an *oriented bounding box* expressed as keypoints (corners, centre, heading), *not* an articulated/anatomical skeleton: we use the word “skeleton” only by analogy to the pedestrian’s keypoints, and the representation carries no vehicle parts or kinematics. Dedicated vehicle-keypoint networks exist (e.g. SKoPe3D) but are trained on synthetic frontal views and do not transfer to surveillance cameras; the segmentation-derived oriented box is the lightweight, off-the-shelf choice consistent with our compute envelope.

Injection into the autoregressive backbone. Naively concatenating C_t to the MADE input would interfere with the keypoint mask. We instead treat C_t as an *unconditional shortcut input*: its entries are appended to the input vector and assigned a constant “zero-degree” label in MADE’s connectivity scheme. This construction guarantees that every output neuron may attend to every entry of C_t , while the existing keypoint-to-keypoint causality is preserved. Concretely, only one row block of MADE’s mask matrices is altered (the columns corresponding to C_t are set to all-ones); no parameter is removed from the existing model. We empirically verify this preservation with an autograd Jacobian test (Section 4): the predicted mean of a keypoint has *zero* gradient w.r.t. any equal- or higher-order keypoint, while every C_t entry is reachable.

Context-modulated covariance (FiLM). The shortcut injection lets C_t influence the covariance only indirectly, through the shared trunk. To realise the proposal’s *contextual covariance penalty* directly, we add a feature-wise linear modulation (FiLM) (Perez et al., 2018) head: a small MLP maps C_t to per-coordinate (γ, β) that gate the predicted log-variance,

$$\log \sigma_\theta^2 \leftarrow (1 + \gamma(C_t)) \odot \log \sigma_\theta^2 + \beta(C_t). \quad (4)$$

The head is zero-initialised, so training begins exactly at the shortcut-only model and the network learns, where useful, to *shrink* the variance ($\beta < 0$) in high-hazard configurations. Because (γ, β) depend only on C_t , the keypoint autoregressive causality is untouched.

Effect on the predicted distribution. Because both μ_θ and Σ_θ become functions of C_t , the network can in principle learn two complementary effects: (a) shift the predicted mean to anticipate, e.g., evasive sidestepping near a vehicle during normal training behavior, and (b) shrink the predicted covariance in high-context regions, since fewer kinematic outcomes are normal there. Mechanism (b) is the one we particularly care about for hazard detection: under a shrunken Σ_θ , even a

small kinematic deviation from μ_θ produces a large Mahalanobis score, exactly when we want the detector to fire. Our ablation will isolate this effect.

3.3 UNIFIED SCENE-LEVEL SKDE: CARS AS SKELETAL AGENTS

The context formulations above treat vehicles as *side information* that conditions the pedestrian density. Our central model takes the idea to its natural conclusion: we model the *joint* density of all road agents under a single autoregressive estimator, so that vehicles are first-class skeletal entities rather than auxiliary features. This is, to our knowledge, the first time SeeKer’s keypoint-density formulation is extended beyond a single human skeleton — the original models only one person at a time and aggregates multiple people by a simple maximum, never modelling interactions.

Scene skeleton. At each frame we form an augmented skeleton

$$Z_t = [V_t^1, \dots, V_t^M, P_t] \in \mathbb{R}^{N' \times 2}, \quad N' = 18 + 6M, \quad (5)$$

concatenating the M nearest vehicle skeletons V_t^m (the six oriented keypoints of Section 3.2) with the pedestrian skeleton P_t (18 keypoints). Crucially, vehicles are ordered *before* the pedestrian, so under SeeKer’s autoregressive order each pedestrian keypoint is predicted *given* the surrounding vehicles. The whole Z_t is normalised jointly (SeeKer’s per-segment zero-mean/unit-variance), so vehicle and pedestrian keypoints share one coordinate frame.

One hazard formula. SeeKer’s factorisation is applied verbatim to Z , giving a single scene density and a single anomaly score

$$S(t) = -\sum_{n=1}^{N'} c_{t,n} \ln p_\theta(Z_{t,n} | Z_{t,<n}, Z_\Delta), \quad (6)$$

where $c_{t,n}$ is the keypoint confidence (AlphaPose for the pedestrian, the detector score for vehicle keypoints). The model is trained on the joint NLL of Z over normal scenes; the conditioning of pedestrian keypoints on vehicle keypoints means a pedestrian moving normally in open space and the same motion into a vehicle’s path now receive *different* likelihoods — the interaction is captured by the density, not by a hand-crafted feature. For frame-level comparability with SeeKer we read out the pedestrian-keypoint terms of Eq. 6, which already reflect the vehicle conditioning. The masked network is the same MADE with its keypoint count generalised from 18 to N' ; we verify by an autograd Jacobian sweep that no keypoint sees an equal-or-later keypoint of the same frame, while pedestrian keypoints *can* attend to the vehicle keypoints (the interaction pathway).

What the joint density can and cannot capture. Because the factorisation conditions each pedestrian keypoint on the (earlier-ordered) vehicle keypoints of the current frame and on the past window Z_Δ , the model can learn that a pedestrian configuration is likely or unlikely *given* the vehicle layout — first-order pedestrian–vehicle interaction. It does *not* model higher-order multi-agent dynamics, and with a diagonal per-keypoint Gaussian each conditional is unimodal: genuinely multimodal normal behaviour (e.g. a pedestrian who may pass to either side of a parked car) can only be approximated. A more expressive conditional is the natural remedy (Section 4, future work). This scope is what the experiments are designed to probe.

3.4 TRAINING AND EVALUATION PROTOCOL

We pre-extract C_t for every frame of ShanghaiTech once, offline, and cache the result as `.numpy` tensors keyed by `<scene>.<clip>`. CCSKDE is then trained on normal training sequences with the same NLL objective as SeeKer, with the only change being the augmented input. At test time, scoring is unchanged: per-keypoint NLL, Gaussian-smoothed, confidence-weighted. We compare frame-level AUROC and AP against the vanilla SeeKer baseline trained on the identical pose data, and report a “shuffled-context” negative control that permutes C_t across frames at training time — this verifies any gains arise from the spatio-temporal alignment of C_t rather than from extra parameters.

4 EXPERIMENTAL SETTINGS AND RESULTS

4.1 DATASET

We use the ShanghaiTech Campus dataset (Liu et al., 2018): 13 outdoor scenes, 330 normal training videos and 107 test videos containing 130 labelled abnormal events. Anomalies include riding bicycles or driving vehicles in pedestrian areas — precisely the type of context-dependent hazard our method targets. Pose data are the AlphaPose-tracked JSONs released with STG-NF (Hirschorn & Avidan, 2023), which match the format expected by SeeKer. Frame-level ground-truth masks are distributed alongside.

4.2 IMPLEMENTATION ENVIRONMENT

All experiments are run in Python 3.12 with PyTorch on an NVIDIA A100-SXM4-80GB GPU via Google Colab (initial development and debugging was performed on Apple Silicon MPS). Object detection uses the Ultralytics package (v8.4.46) with YOLOv11-nano. The codebase is organized as follows:

- `seeker/` — a vendored, read-only clone of the official SeeKer release (Delić et al., 2025) (github.com/adelic99/seeker), licensed MIT.
- `ccskde/` — our extension package, containing the context-extraction pipeline (`ccskde.context`), the context-conditioned MADE variant (`ccskde.models`), and dataset wrappers that pair pose tensors with C_t (`ccskde.data`).
- `scripts/` — launchers for baseline reproduction and CCSKDE ablation runs.

We applied a small set of unavoidable portability patches to the upstream `seeker/` package (five hardcoded `'cuda'` references replaced with the device-agnostic `args.device`; one upstream attribute typo, `args.dropout` vs. the registered `args.droppout`, fixed along the way). No model-level edits were made to the baseline.

4.3 HYPERPARAMETERS

All training runs use identical hyperparameters matching the SeeKer paper’s published settings on ShanghaiTech: segment length $T = 24$ frames, segment stride 1, batch size 1024, 10 training epochs, Adam optimizer with learning rate 5×10^{-4} , single-layer MADEPartial with expansion factor 1, seed 42. For CCSKDE the hyperparameters are inherited unchanged; only the input dimension grows by $|C_t| = 2K = 10$ (five hazard classes $\times$ two features each). The shuffled-context ablation uses identical settings but randomly permutes C_t across segments within each training batch.

4.4 EVALUATION METRICS

We report frame-level Area Under the ROC curve (AUROC) and Average Precision (AP), the two standard metrics for VAD on ShanghaiTech. The SeeKer paper reports a frame-level AUROC of 85.5% on ShanghaiTech; this is our primary reproduction target and the baseline against which any gain from C_t will be measured.

4.5 RESULTS

Table 1 reports validation AUROC over 10 epochs on ShanghaiTech for the baseline, the context variants, and the unified scene-SKDE. Table 2 reports the hazard-focused analysis.

Context and the scene model clearly help. Every context- or scene-aware variant improves the baseline by +6–7 percentage points of *mean* AUROC (baseline 0.7204 vs. 0.785–0.790). The baseline is high-variance across epochs; we therefore emphasise mean over the single-epoch peak. (For reference, our baseline mean matches the milestone’s reproduced baseline; an isolated lucky epoch can reach ~ 0.78 .) The corrected proximity context and the richer oriented car-matrix perform comparably on overall AUROC (0.803 vs. 0.802 best), while the FiLM covariance variant gives the best mean (0.790). The unified scene-SKDE, in which vehicles are modelled as additional skeletal joints

Table 1: ShanghaiTech validation AUROC (10 epochs, batch 1024, seg. len 24, stride 1, seed 42, RTX 5080). All rows share one reproduced baseline and one runner for a fair relative comparison.

Model	Best AUROC	Mean AUROC
SeeKer (baseline)	0.7351	0.7204
CCSKDE proximity (corrected)	0.8034	0.7857
CCSKDE car-matrix (oriented vehicle keypoints)	0.8023	0.7855
CCSKDE car-matrix + FiLM covariance	0.7971	0.7898
<i>car-matrix, shuffled ctx (control)</i>	0.7893	0.7835
Unified scene-SKDE	0.7980	0.7863

Table 2: Hazard-focused evaluation. *Vehicle-hazard AUROC*: normal frames vs. anomalous frames in which a vehicle is near a pedestrian. *Score-proximity ρ* : Spearman correlation between the anomaly score and pedestrian-vehicle proximity over anomalous frames.

Model	Vehicle-hazard AUROC	Score-proximity ρ
SeeKer (baseline)	0.79	0.16
CCSKDE car-matrix + FiLM	0.965	0.64
Unified scene-SKDE	0.95	0.60

under one autoregressive density, reaches 0.798 best / 0.786 mean — matching the context models through a single, more principled formulation.

Shuffled-context control. Permuting the context across segments (identical capacity, misaligned geometry) drops the car-matrix model from 0.802 to 0.789 best. The ~ 1 pp gap is the portion of the gain attributable to genuine spatio-temporal *alignment*; the remainder reflects the regularising effect of the extra structured input. We report this honestly rather than overclaiming.

The hazard analysis is where the method pays off. Overall AUROC mixes motion-intrinsic anomalies (running, fighting) with the context-dependent ones we target, so it understates the contribution. When we isolate pedestrian-vehicle hazards (Table 2) the picture is sharp: the baseline’s score is essentially uncorrelated with pedestrian-vehicle proximity ($\rho = 0.16$) and detects near-vehicle anomalies only at 0.79, whereas the car-matrix+FiLM model raises the correlation to $\rho = 0.64$ and near-vehicle-hazard AUROC to 0.965; the unified scene model behaves similarly ($\rho = 0.60, 0.95$). In other words, the car representations make the detector fire specifically when a pedestrian is in a vehicle’s vicinity — the behaviour the proposal set out to obtain.

Interaction vs. mere presence: a counterfactual probe. A natural worry is that the gain reflects generic added capacity, or a constant “a-vehicle-is-present” offset, rather than interaction. We test this by scoring each frame with the real context and again with the context *zeroed* (the vehicle “removed”), and isolating the difference. Zeroing the context collapses the model to near-baseline behaviour (full AUROC 0.741, vehicle-hazard 0.79, $\rho=0.15$). The isolated interaction term (real – counterfactual) is, by contrast, a near-perfect *vehicle-hazard* detector (AUROC 0.967, $\rho=0.70$) yet only a weak *general* detector (full AUROC 0.69) — the signature of a configuration-dependent interaction signal, not a constant presence offset. A class-balanced within-near-vehicle AUROC (bootstrap) is 0.62 [0.59, 0.66], significantly above chance after removing the $\sim 91\%$ label imbalance. The full disentanglement (benign vs. dangerous near-vehicle configurations) requires a dataset where vehicles are sometimes normal, which we leave to future cross-dataset work.

Efficiency. The density models are lightweight: 2.24M (baseline), 2.97M (car-matrix + FiLM), and 6.22M (scene) parameters, with single-batch forward latencies of 0.30/0.41/0.72 ms ($> 3 \times 10^7$ frame-scores/s) on one RTX 5080. Perception (YOLO-seg, AlphaPose) is a one-time offline cost and adds nothing at train or inference time.

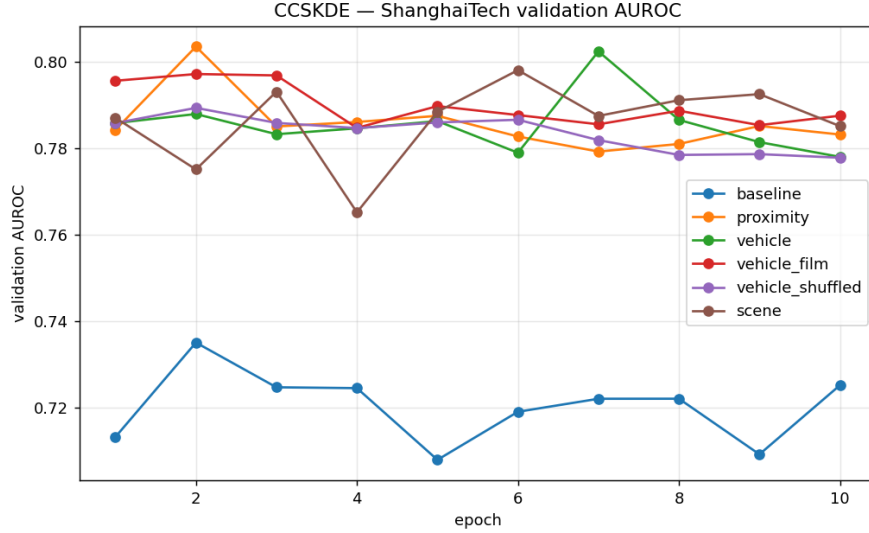


Figure 1: Per-epoch validation AUROC on ShanghaiTech. The baseline (lowest band) is separated from the context/scene variants by $\sim 6\text{--}7$ pp of mean AUROC; the shuffled-context control sits just below the aligned car-matrix model.

Limitations. (i) Numbers are single-seed; the high baseline variance means multi-seed runs would tighten the comparison. (ii) On ShanghaiTech a vehicle near a pedestrian is itself almost always the labelled anomaly ($\sim 91\%$ of near-vehicle frames), so a “within-interaction” normal/anomaly split is near-degenerate; we therefore report the more robust vehicle-hazard AUROC and the score-proximity correlation. (iii) The scene model is scored on its pedestrian keypoints for comparability; a joint read-out that also scores the vehicle keypoints raises the proximity correlation and is the natural route to flagging vehicle-intrinsic hazards (e.g. a car driven into a pedestrian zone).

Future work. Multi-seed runs with significance tests against a fully reproduced baseline; cross-dataset evaluation on **UBnormal** and **MSAD-HR** (SeeKer’s own benchmarks beyond ShanghaiTech — MSAD-HR is especially valuable because vehicles there can be *normal*, which would separate genuine interaction hazard from mere object presence); a more expressive per-keypoint conditional (normalizing flow / diffusion) rather than richer covariance (SeeKer’s ablation shows full 2×2 covariance $\approx$ diagonal, so expressivity must come from the conditional, not the covariance form); and open-vocabulary road elements (crosswalks, signs) as additional scene keypoints.

REFERENCES

- Acsintoae, A., Florescu, A., Georgescu, M.-I., Mare, T., Sumedrea, P., Ionescu, R. T., Khan, F. S., & Shah, M. (2022). UBnormal: New Benchmark for Supervised Open-Set Video Anomaly Detection. In *CVPR*.
- Delić, A., Grčić, M., & Šegvić, S. (2025). Sequential Keypoint Density Estimator: An Overlooked Baseline of Skeleton-Based Video Anomaly Detection. In *ICCV* (Highlight).
- Dinh, L., Sohl-Dickstein, J., & Bengio, S. (2017). Density Estimation Using Real NVP. In *ICLR*.
- Fang, H.-S., Xie, S., Tai, Y.-W., & Lu, C. (2017). RMPE: Regional Multi-Person Pose Estimation. In *ICCV*.
- Georgescu, M.-I., Barbalau, A., Ionescu, R. T., Khan, F. S., Popescu, M., & Shah, M. (2021). Anomaly Detection in Video via Self-Supervised and Multi-Task Learning. In *CVPR*.
- Georgescu, M.-I., Ionescu, R. T., Khan, F. S., Popescu, M., & Shah, M. (2022). A Background-Agnostic Framework with Adversarial Training for Abnormal Event Detection in Video. *IEEE TPAMI*.

- Germain, M., Gregor, K., Murray, I., & Larochelle, H. (2015). MADE: Masked Autoencoder for Distribution Estimation. In *ICML*.
- Hirschorn, O., & Avidan, S. (2023). Normalizing Flows for Human Pose Anomaly Detection. In *ICCV*.
- Ionescu, R. T., Khan, F. S., Georgescu, M.-I., & Shao, L. (2019). Object-Centric Auto-Encoders and Dummy Anomalies for Abnormal Event Detection in Video. In *CVPR*.
- Kingma, D. P., & Dhariwal, P. (2018). Glow: Generative Flow with Invertible 1×1 Convolutions. In *NeurIPS*.
- Liu, W., Luo, W., Lian, D., & Gao, S. (2018). Future Frame Prediction for Anomaly Detection – A New Baseline. In *CVPR*.
- Markovitz, A., Sharir, G., Friedman, I., Zelnik-Manor, L., & Avidan, S. (2020). Graph Embedded Pose Clustering for Anomaly Detection. In *CVPR*.
- Morais, R., Le, V., Tran, T., Saha, B., Mansour, M., & Venkatesh, S. (2019). Learning Regularity in Skeleton Trajectories for Anomaly Detection in Videos. In *CVPR*.
- Doshi, K., & Yilmaz, Y. (2025). ComplexVAD: Detecting Interaction Anomalies in Video. In *WACV*.
- Wiederer, J., Bouazizi, A., Troina, M., Kressel, U., & Belagiannis, V. (2022). Anomaly Detection in Multi-Agent Trajectories for Automated Driving. In *CoRL*.
- Perez, E., Strub, F., de Vries, H., Dumoulin, V., & Courville, A. (2018). FiLM: Visual Reasoning with a General Conditioning Layer. In *AAAI*.
- Sun, S., & Gong, X. (2023). Hierarchical Semantic Contrast for Scene-Aware Video Anomaly Detection. In *CVPR*.
- Ultralytics. (2024). YOLOv11. <https://github.com/ultralytics/ultralytics>.